Multiple Choice:

1. A. 1.12 MPa

$$s = \frac{Pd}{2t}$$

$$140 \times 10^6 \frac{N}{m^2} = \frac{P(0.5\text{ m})}{2(0.02\text{ m})}$$

$$P = 1.12\text{ MPa} \rightarrow \text{Ans}$$

Reference: GEAS 05-05 – Pressure Vessels

2. B. 5 mm

$$s = \frac{P}{A}$$

$$140 \times 10^6 = \frac{2000}{\frac{p}{4}d^2}$$

$$d = 4.26\text{ mm} \rightarrow 1$$

$$\delta = \frac{PL}{AE}$$

$$0.005 = \frac{2000(10)}{\left(\frac{\pi}{4}d^2\right)(200 \times 10^9)}$$

$$d = 5.05\text{ mm} \rightarrow 2$$

$$\therefore\ d = 5.05\text{ mm} \rightarrow \text{Ans}$$

Reference: GEAS 05-02 – Stress-Strain Diagram

3. C. 118 mm

$$q = \frac{TL}{JG}$$

$$3° \times \frac{p}{180°} = \frac{(14000)(6)}{\frac{p}{32}d^4(83 \times 10^9)}$$

$$d = 118\text{ mm} \rightarrow \text{Ans}$$

Reference: GEAS 05-06 - Torsion

4. A. 571 mm

$$d = \frac{64PR^3n}{Gd^4}$$

$$d = \frac{64(1500\text{ N})(0.10\text{ m})^3(25)}{\left(83 \times 10^9 \frac{N}{m^2}\right)(0.015\text{ m})^4}$$

$$d = 571\text{ mm} \rightarrow \text{Ans}$$

Reference: GEAS 05-07 – Helical Springs

5. B. 47 MPa

$$s_{\text{thermal}} = aEDT$$

$$s_{\text{thermal}} = (11.7 \times 10^{-6}/°C)(200 \times 10^9\text{ Pa})(20°C)$$

$$s_{\text{thermal}} = 46.8\text{ MPa} \rightarrow \text{Ans}$$

Reference: GEAS 045-05 – Thermaal Stress

6. C. 3.3 N

$$d = \frac{PL}{AE}$$

$$0.0015\text{ m} = \frac{P(100\text{ m} - 0.0015\text{ m})}{(0.010\text{m})(0.00011\text{m})(200 \times 10^9)}$$

$$P = 3.3\text{ N} \rightarrow \text{Ans}$$

Reference: GEAS 05-02 – Stress-Strain Diagram

7. C. 50 MPa

$$s = \frac{pD}{2t} = \frac{(5\text{ MPa})(200\text{ mm})}{2(10\text{ mm})} = 50\text{ MPa} \rightarrow \text{Ans}$$

Reference: GEAS 05-05 – Pressure Vessels

8. D. 183 mm

Compressive stress is a type of Normal or Axial stress. This means that the area should be in a plane perpendicular to the applied force.

$$s = \frac{P}{A}$$

$$50 \times 10^6 \frac{N}{m^2} = \frac{250{,}000\text{ N}}{\frac{p}{4}(0.2\text{ m}^2) - \frac{p}{4}d^2}$$

$$d = 183.4\text{ mm} \rightarrow \text{Ans}$$

Reference: GEAS 05-01 – Simple Stress

9. C. 42 mm

$$q = \frac{TL}{JG}$$

$$4° \times \frac{p}{180°} = \frac{(18{,}000\text{ Nm})(10\text{ m})}{\left[\frac{p}{32}d^4\right]\left(83 \times 10^9 \frac{N}{m^2}\right)}$$

$$d = 41.8\text{ mm}$$

Reference: GEAS 05-06 - Torsion

10. D. 50 mm

$$t = \frac{16T}{pd^3}$$

$$60 \times 10^6 \frac{N}{m^2} = \frac{16T}{pd^3} \rightarrow \text{eq 1}$$

$$q = \frac{TL}{JG}$$

$$5° \times \frac{p}{180°} = \frac{T(3\text{ m})}{\left[\frac{p}{32}d^4\right]\left(83 \times 10^9 \frac{N}{m^2}\right)} \rightarrow \text{eq 2}$$

Two equations, two unknowns: d = 49.7 m

Reference: GEAS 05-06 - Torsion

11. A. 226 mm

$$d = \frac{64PR^3n}{Gd^4}$$

$$t = \frac{16PR}{pd^3}\left[\frac{4m - 1}{4m - 4} + \frac{0.615}{m}\right]$$

$$m = \frac{D}{d} = \frac{160}{20} = 8$$

$$140\text{x}10^6 = \frac{16P(0.08)}{p(0.02)^3}\left[\frac{4(8) - 1}{4(8) - 4} + \frac{0.615}{8}\right]$$

$$P = 2322\text{ N}$$

$$d = \frac{64(2322)(0.08)^3(20)}{(42\text{x}10^9)(0.02)^4}$$

$$d = 226.45\text{ mm} \rightarrow \text{Ans}$$

Reference: GEAS 05-07 – Helical Springs

12. D. 49.7 mm

$$q = \frac{TL}{JG}$$

$$5° \times \frac{p}{180°} = \frac{T(5\text{ m})}{\frac{p}{32}d^4\left(83 \times 10^9 \frac{N}{m^2}\right)}$$

$$T = 237.03 \times 10^6 d^4$$

$$t = \frac{16T}{pd^3}$$

$$60 \times 10^6 \frac{N}{m^2} = \frac{16(237.03 \times 10^6 d^4)}{pd^3}$$

$$d = 49.7\text{ mm} \rightarrow \text{Ans}$$

Reference: GEAS 05-06 - Torsion

13. D. 0.092 mm

The total elongation of the rod is the sum of the elongation caused by its own weight and the elongation caused by the 10 kN tensile load. Solve for the elongation caused by its own weight:

$$\delta_{weight} = \frac{\rho g L^2}{2E} = \frac{mgL}{2AE}$$

$$\delta_{weight} = \frac{(7850)(9.81)(20)^2}{2\left(200\times10^9\right)} = 0.078 \text{ mm}$$

Solve for the elongation caused by the tensile load:

$$\delta_{load} = \frac{PL}{AE} = \frac{(10{,}000)(20)}{\frac{\pi}{4}(0.3)^2\left(200\times10^9\right)}$$

$$\delta_{load} = 0.014 \text{ mm}$$

Get the total elongation:

$$d_{total} = 0.078 \text{ mm} + 0.014 \text{ mm}$$

$$d_{total} = 0.092 \text{ mm} \rightarrow \text{Ans}$$

Reference: GEAS 05-02 – Stress-Strain Diagram

14. A. 114 mm

$$q = \frac{TL}{JG}$$

$$3^o\left(\frac{p}{180^o}\right) = \frac{(12\text{kNm})(6\text{m})}{\frac{p}{32}d^4(83\,\text{GPa})}$$

$$d = 114 \text{ mm} \rightarrow \text{Ans}$$

Reference: GEAS 05-06 - Torsion

15. D. 192 kN

$$d = \frac{PL}{AE};\quad P = \frac{dAE}{L}$$

$$P_T = \frac{dAE}{L}_{steel} + \frac{dAE}{L}_{iron}$$

$$= \frac{0.0008\left(\frac{p}{4}(0.05)^2\right)\left(200\times10^9\right)}{2} + \frac{0.0008\left(\frac{p}{4}\left(0.06^2 - 0.05^2\right)\right)\left(100\times10^9\right)}{2}$$

$$P_T = 192 \text{ kN} \rightarrow \text{Ans}$$

Reference: GEAS 05-02 – Stress-Strain Diagram

16. C. 7.96 x 10^7 N/m^2

$$\tau_{max} = \frac{16T}{\pi d^3} = \frac{16(1\,\text{kNm})}{\pi(0.04\text{m})^3} = 79.6 \text{ MPa}$$

Reference: GEAS 05-06 - Torsion

17. A. 8.56°

$$\theta = \frac{TL}{JG} = \frac{1\,\text{kNm}(3\text{m})}{\frac{\pi}{32}(0.04\text{m})^4(80\text{ GPa})}$$

$$\theta = 0.149\,\text{rad}\times\frac{180^o}{\pi} = 8.55^o \rightarrow \text{Ans}$$

Reference: GEAS 05-06 - Torsion

18. B.104 mm

For mild steel, $t_{max} = \frac{16T}{pd^3}$

For alloy steel,

$$t_{max} = 1.22t_{max(mild\ steel)}$$

$$t_{max} = \frac{16TD}{p\left(D^4 - d^4\right)}$$

But T should be,

$$P = 2pfT$$

$$1.2P = 2p(1.06f)T$$

$$\backslash\quad T = 1.13T$$

Therefore,

$$1.22\left(\frac{16T}{pd^3}\right) = \frac{16(1.13T)(D)}{p\left(D^4 - d^4\right)}$$

$$1.22\left(\frac{16}{p(0.2\text{m})^3}\right) = \frac{16(1.13)(0.2\text{m})}{p\left[(0.2\text{m})^4 - d^4\right]}$$

$$d = 104 \text{ mm} \rightarrow \text{Ans}$$

Reference: GEAS 05-06 - Torsion

19. A. 297 MPa

$$\tau_{max} = \frac{16T}{\pi d^3} = \frac{16(20\text{kNm})}{\pi(0.07\text{m})^3} = 297 \text{ MPa}$$

Reference: GEAS 05-06 - Torsion

20. D. 4.86°

$$\theta = \frac{TL}{JG}$$

$$\frac{\theta}{L} = \frac{T}{JG} = \frac{20\text{kNm}}{\frac{\pi}{32}(0.07\text{m})^4\left(10^5\text{MPa}\right)}$$

$$\frac{\theta}{L} = 0.085\,\text{rad}\times\frac{180^o}{\pi} = 4.86^o \rightarrow \text{Ans}$$

Reference: GEAS 05-06 - Torsion

21. A. 500 MPa

$$s = \frac{P}{A} = \frac{10\text{kN}}{20\text{mm}^2\times\left(\frac{10^{-3}\text{m}}{1\,\text{mm}}\right)^2}$$

$$s = 500 \text{ MPa} \rightarrow \text{Ans}$$

Reference: GEAS 05-01 – Simple Stress

22. A. 500 MPa

$$s = \frac{P}{A} = \frac{10\text{kN}}{20\text{mm}^2\times\left(\frac{10^{-3}\text{m}}{1\,\text{mm}}\right)^2}$$

$$s = 500 \text{ MPa} \rightarrow \text{Ans}$$

Reference: GEAS 05-01 – Simple Stress

23. D. If a material expands freely due to heating, it will develop thermal stresses.

24. A. EαT

25. C. 17.5 mm

$$s_c = \frac{P}{A}$$

$$800 \text{ MPa} = \frac{P}{\frac{p}{4}d^2}$$

$$P = 800 \text{ MPa}\left(\frac{p}{4}d^2\right) \rightarrow 1$$

$$t = \frac{P}{A}$$

$$350 \text{ MPa} = \frac{800 \text{ MPa}\left(\frac{p}{4}d^2\right)}{pd(0.010\text{m})}$$

$$d = 17.5 \text{ mm} \rightarrow \text{Ans}$$

Reference: GEAS 05-01 – Simple Stress